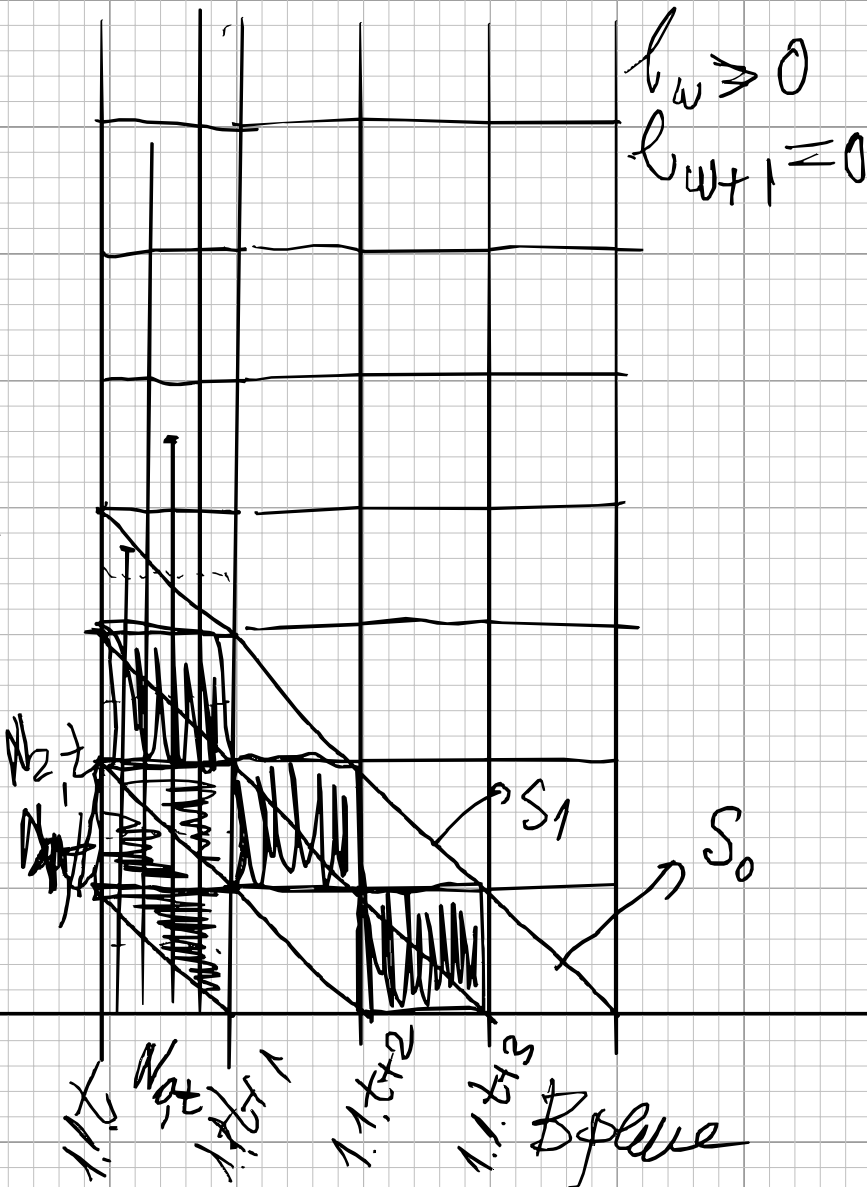


Bar graph of N



$$l_w \geq 0$$

$$l_{w+1} = 0$$

$$\frac{N_{x+1,t}}{N_{x,t}} = p_x = \frac{l_{x+1}}{l_x}$$

$$q_x = 1 - p_x$$

$$l_0 = 100\,000$$

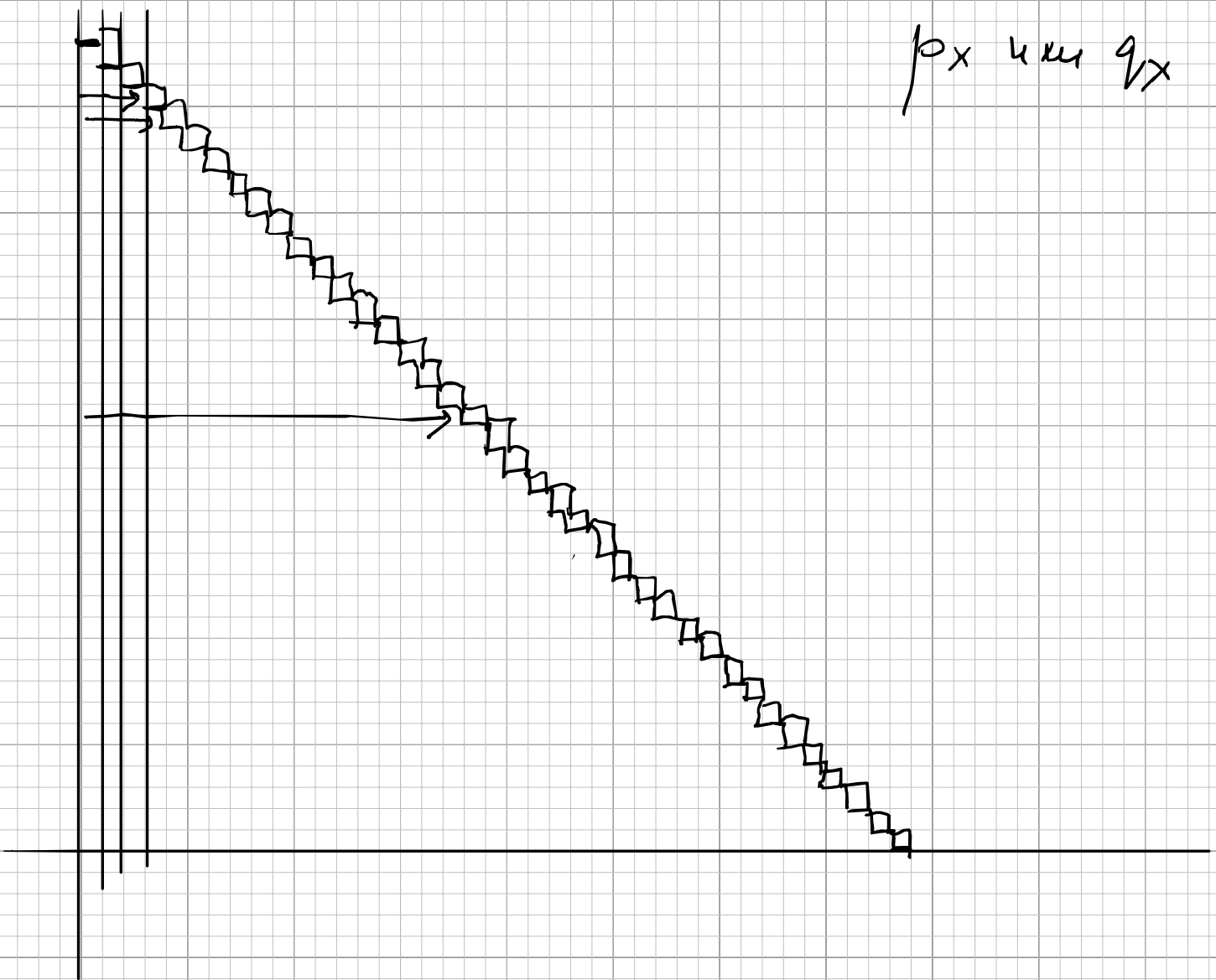
$$\frac{d_x}{l_x} = q_x \rightarrow d_x = l_x q_x$$

$$\frac{l_x + l_{x+1}}{2} = l_x$$

$$\sum_{i=x}^{\infty} L_i = T_x$$

$$e_x = \frac{T_x}{l_x}$$

ρ_x и q_x



$$\begin{pmatrix} p_{3,3} & p_{3,5} & p_{3,y} \\ p_{5,3} & p_{5,5} & p_{5,y} \\ 0 & 0 & 1 \end{pmatrix}^*$$

$$\pi_{x,3} = \frac{S_{x,3}}{S_x}$$

$$\begin{pmatrix} p_{3,3} & p_{3,5} & p_{3,y} \\ 0 & p_{5,5} & p_{5,y} \\ 0 & 0 & 1 \end{pmatrix}^*$$

$$l_x \cdot \pi_{x,3} = l'_x$$

